MTH240

**Toronto Metropolitan University**

Family Name:_____**SOLUTIONS**_____Given Name:___ MTH240 Midterm Q1 ___

TMU Email:_____

Student ID # (ONLY LAST 3 DIGITS):_____

Signature:_____

Date/time: May 27, 2024, 6:10 – 8:10 pm

**Section (Circle One)**

| 01 | 02 | 03 | 04 |
|---|---|---|---|

**Instructions**

1. Calculators, notes, and other aids are **not** allowed.
2. **Tests written in pencil are not eligible for remarking.**
3. Marks (out of) are shown in a box to the left.
4. Answer all questions in this booklet. If you need extra room, use the last two pages, clearly indicating where your answer continues.
5. In every question, show your work, presented clearly and in the correct order. **Unjustified answers will be given little or no credit.**
6. Cross out all irrelevant or incorrect work, as marks may be deducted for work which is misleading, irrelevant or incorrect.
7. Keep your TMU ID displayed on your desk.
8. Do not separate the sheets,
9. Make sure your test paper is complete; there are 7 questions on 10 pages (including this one). The final two pages are given for extra space and do NOT contain any questions.

| Question Number | Score |
|---|---|
| **Q1** | **/6** |
| **Q2** | **/8** |
| **Q3** | **/6** |
| **Q4** | **/5** |
| **Q5** | **/3** |
| **Q6** | **/6** |
| **Q7** | **/6** |
| **Total** | **/40** |

6 Marks

1. Determine whether the following integral is convergent or divergent. Evaluate if it is convergent. Show your work.

$$\int_1^{\infty} \frac{\ln(x)}{x^3}\,dx$$

$$\int_1^{\infty} \frac{\ln(x)}{x^3}\,dx = lim_{t\to\infty}\int_1^{t} \frac{\ln(x)}{x^3}\,dx\ ,\ \int \frac{\ln(x)}{x^3}\,dx = ?$$

$$\int u\cdot dv = u\cdot v - \int v\cdot du$$

$$u = \ln(x) \qquad dv = \frac{1}{x^3} = x^{-3}$$

$$du = \frac{1}{x}dx \qquad v = \int x^{-3}dx = \frac{x^{-3+1}}{(-3+1)} = \frac{x^{-2}}{-2} = \frac{\left(\frac{1}{x^2}\right)}{-2} = \frac{1}{x^2}\cdot -\frac{1}{2} = -\frac{1}{2x^2}$$

$$\int \frac{\ln(x)}{x^3}dx = -\frac{\ln(x)}{2x^2} - \int -\frac{1}{2x^2}\cdot\frac{1}{x}dx = -\frac{\ln(x)}{2x^2} + \frac{1}{2}\int \frac{1}{x^3}dx = -\frac{\ln(x)}{2x^2} + \frac{1}{2}\cdot -\frac{1}{2x^2} = -\frac{\ln(x)}{2x^2} - \frac{1}{4x^2} + C$$

$$\int_1^{\infty} \frac{\ln(x)}{x^3}dx = lim_{t\to\infty}\int_1^{t} \frac{\ln(x)}{x^3}dx = lim_{t\to\infty}\left[-\frac{\ln(x)}{2x^2} - \frac{1}{4x^2}\right]_1^t = -\frac{\ln(t)}{2t^2} + \frac{1}{2t^2} - \left(-\frac{\ln(1)}{2(1)^2} - \frac{1}{4(1)^2}\right)$$

$$= -\frac{\ln(t)}{2t^2} + \frac{1}{2t^2} + \frac{\ln(1)}{2(1)^2} + \frac{1}{4(1)^2} = \frac{\infty}{\infty} + \frac{1}{\infty} + 0 + \frac{1}{4} = \frac{\infty}{\infty} + \frac{1}{4}$$

L'Hopital:

$$= lim_{t\to\infty} -\frac{\ln(t)}{2t^2} = \frac{\infty}{\infty}: = lim_{t\to\infty}\frac{\frac{d}{dx}(\ln(t))}{\frac{d}{dx}(-2t^2)} = lim_{t\to\infty}\frac{\frac{1}{t}}{-4t} = lim_{t\to\infty}\frac{1}{t}\cdot\frac{1}{-4t} = lim_{t\to\infty}\frac{1}{-4t^2} = \frac{1}{\infty} = 0$$

$$\int_1^{\infty} \frac{\ln(x)}{x^3}dx = lim_{t\to\infty}\int_1^{t} \frac{\ln(x)}{x^3}dx = lim_{t\to\infty}\left[-\frac{\ln(x)}{2x^2} - \frac{1}{4x^2}\right]_1^t = 0 + \frac{1}{4} = \frac{1}{4}$$

Therefore, the following integral is convergent.

8 Marks

2. Evaluate the following integral. Write your final answer without trigonometric expressions.

$$\int \frac{dx}{x^4\sqrt{x^2-4}}$$

$\sqrt{x^2 - 4} \Rightarrow \sqrt{x^2 - a^2} : x = a\,sec\,(\theta); \qquad a^2 = 4 \Rightarrow a = \sqrt{4} = 2$

$x = 2\,sec\,(\theta) \;\;,\;\; dx = 2sec(\theta)tan(\theta)\,d\theta$

$\sqrt{x^2 - 4}$, where $x = 2\,sec\,(\theta)$:

$\sqrt{(2\,sec\,(\theta))^2 - 4} = \sqrt{4\,sec^2(\theta) - 4} = \sqrt{4(\,sec^2(\theta) - 1)} = \sqrt{4\,tan^2(\theta)} = 2\,tan(\theta)$

$\int \frac{dx}{x^4\sqrt{x^2-4}} = \int \frac{2\,sec(\theta)\,tan(\theta)\,d\theta}{(2\,sec(\theta))^4 2\,tan(\theta)} = \int \frac{2\,sec(\theta)\,tan(\theta)\,d\theta}{16\,sec^4(\theta)\;2\,tan(\theta)} = \int \frac{1}{16\,sec^3(\theta)}d\theta = \int \frac{1}{16}\,cos^3(\theta)d\theta$

$= \frac{1}{16}\int cos^3(\theta)d\theta \qquad \int cos^3(\theta)d\theta = \;?$

$\int cos^3(\theta)d\theta = \int cos^2(\theta)\,cos(\theta)\,d\theta \qquad\qquad u = sin(\theta)\,,\;\; du = cos(\theta)\,d\theta$

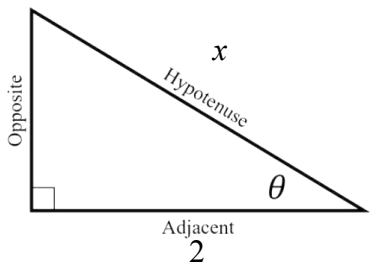
$\int cos^2(\theta)\,cos(\theta)\,d\theta = \int (1 - sin^2(\theta))\,cos(\theta)\,d\theta = \int 1 - u^2\,du$

$\int 1 - u^2\,du = u - \frac{u^3}{3} = sin(\theta) - \frac{sin^3(\theta)}{3} + C\,, \quad sin(\theta) = \;?$

$S\frac{O}{H}C\frac{A}{H}T\frac{O}{A} \quad , \qquad a^2 + b^2 = c^2$

$x = 2\,sec\,(\theta) \rightarrow\ sec\,(\theta) = \frac{x}{2} = \frac{H}{A}$

O = ?




$$O^2 + 2^2 = x^2 \Rightarrow O^2 = x^2 - 4 \Rightarrow O = \sqrt{x^2 - 4}$$

$$sin(\theta) = \frac{\sqrt{x^2-4}}{x} \quad , \qquad sin^3(\theta) = \left(\frac{(x^2-4)^{\frac{1}{2}}}{x}\right)^3 = \frac{(x^2-4)^{\frac{3}{2}}}{x^3}$$

$$\frac{1}{16}\int cos^3(\theta)d\theta = \frac{1}{16}\left(\frac{\sqrt{x^2-4}}{x} - \frac{(x^2-4)^{\frac{3}{2}}}{x^3}\right) = \frac{1}{16x}\left(\sqrt{x^2 - 4} - \frac{(x^2-4)^{\frac{3}{2}}}{x^2}\right) + C$$

Therefore, $\int \frac{dx}{x^4\sqrt{x^2-4}} = \frac{1}{16x}\left(\sqrt{x^2 - 4} - \frac{(x^2-4)^{\frac{3}{2}}}{x^2}\right) + C$

6 Marks

3. Solve the following initial value problem. Solve explicitly for *y(x)*.

$$y' + y = \cos\left(e^x\right) \qquad y(\ln \pi) = 2$$

$y' + P(x)y = Q(x)$ , $\mu(x) = e^{\int P(x)\,dx}$ , $P(x) = ?$

$y' + 1 \cdot y = cos\left(e^x\right)$, $P(x) = 1$

$\mu(x) = e^{\int P(x)\,dx} = e^{\int 1\,dx} = e^x$

$e^x\left[y' + y = cos\left(e^x\right)\right] \Rightarrow e^x y' + e^x y = e^x cos\left(e^x\right) \Rightarrow \left[e^x y\right]' = e^x cos\left(e^x\right)$

$\int \left[e^x y\right]' = \int e^x cos\left(e^x\right)$ , $\int e^x cos\left(e^x\right) = ?$

$\int e^x cos\left(e^x\right)$ $\qquad u = e^x \rightarrow du = e^x dx \Rightarrow dx = \frac{1}{e^x} du = \frac{1}{u} du$

$\int e^x cos\left(e^x\right) = \int u\, cos(u) \cdot \frac{1}{u} du = \int cos(u)\, du = sin(u) = sin\left(e^x\right) + C$

$\int \left[e^x y\right]' = \int e^x cos\left(e^x\right) \Rightarrow e^x y = sin\left(e^x\right) + C$, $\qquad y(ln\pi) = 2 \rightarrow (ln\pi,\ 2)$

$e^{ln\pi} \cdot 2 = sin\left(e^{ln\pi}\right) + C \Rightarrow 2\pi = sin(\pi) + C \Rightarrow C = 2\pi - sin(\pi) = 2\pi - 0 = 2\pi$

$e^x y = sin\left(e^x\right) + C \Rightarrow y = \frac{1}{e^x}\left(sin\left(e^x\right) + 2\pi\right)$

Therefore, $y(x) = \frac{1}{e^x}\left(sin\left(e^x\right) + 2\pi\right)$ .

5 Marks

4. Consider the rational function $\frac{4x^2-2x^2+x}{(x+1)(x^2-1)^2(x^2+x+1)^2}$. Write the partial fraction decomposition form of this function. Do NOT evaluate the constants.

$$x^2+x+1=0 \Rightarrow Not\ factorable$$

$$x^2-1=(x-1)(x+1) \Rightarrow (x^2-1)^2=(x-1)^2(x+1)^2$$

$$\frac{4x^2-2x^2+x}{(x+1)(x^2-1)^2(x^2+x+1)^2}=\frac{2x^2+x}{(x+1)(x+1)^2(x-1)^2(x^2+x+1)^2}=\frac{x(2x+1)}{(x+1)^3(x-1)^2(x^2+x+1)^2}=\frac{x(2x+1)}{(x+1)^3(x-1)^2(x^2+x+1)^2}$$

$$=\frac{A}{(x+1)}+\frac{B}{(x+1)^2}+\frac{C}{(x+1)^3}+\frac{D}{(x-1)}+\frac{E}{(x-1)^2}+\frac{Fx+G}{(x^2+x+1)}+\frac{Hx+I}{(x^2+x+1)^2}$$

Therefore, the Partial Fraction Decomposition of $\frac{4x^2-2x^2+x}{(x+1)(x^2-1)^2(x^2+x+1)^2}$

$$=\frac{A}{(x+1)}+\frac{B}{(x+1)^2}+\frac{C}{(x+1)^3}+\frac{D}{(x-1)}+\frac{E}{(x-1)^2}+\frac{Fx+G}{(x^2+x+1)}+\frac{Hx+I}{(x^2+x+1)^2}.$$

3 Marks

5. Determine if the following integral converges or diverges. Justify your answer. [Hint: use comparison theorem in integrals]

$$\int_1^{\infty} \frac{x^2 + \sin^2 x}{x^5}\, dx$$

$$\frac{x^2+\sin^2 x}{x^5} = \frac{x^2}{x^5} + \frac{\sin^2 x}{x^5} = \frac{1}{x^3} + \frac{\sin^2 x}{x^5} < \frac{1}{x^3} + \frac{1}{x^5}$$

If $0 < a < \infty$, then

(a) $\displaystyle\int_a^{\infty} x^{-p}\, dx = \begin{cases} \text{converges to } \dfrac{a^{1-p}}{p-1} & \text{if } p > 1 \\ \text{diverges to } \infty & \text{if } p \leq 1 \end{cases}$

(b) $\displaystyle\int_0^{a} x^{-p}\, dx = \begin{cases} \text{converges to } \dfrac{a^{1-p}}{1-p} & \text{if } p < 1 \\ \text{diverges to } \infty & \text{if } p \geq 1 \end{cases}$

$\frac{1}{x^3}$: $p = 3 > 1$, Convergent

$\frac{1}{x^5}$: $p = 5 > 1$, Convergent

$\frac{1}{x^3} + \frac{1}{x^5} \Rightarrow$ Convergent + Convergent = Convergent

Therefore, the integral converges.

6 Marks

6. Determine whether the following integral is convergent or divergent. Evaluate it if it is convergent. Show your work.

$$\int_0^2 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx$$

$$(x-1)^{\frac{2}{3}}=0 \quad x-1=0 \quad x=1 \quad , \quad x \neq 1$$

$$\int_0^2 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx = \int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx + \int_1^2 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx$$

$$I = I_1 + I_2$$

$$\int_0^2 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx = lim_{t\to 1}\int_0^t \frac{1}{(x-1)^{\frac{2}{3}}}\,dx + lim_{t\to 1}\int_t^2 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx$$

$$I_1 = lim_{t\to 1}\int_0^t \frac{1}{(x-1)^{\frac{2}{3}}}\,dx \quad , \quad \int \frac{1}{(x-1)^{\frac{2}{3}}}dx = \ ?$$

$$\int \frac{1}{(x-1)^{\frac{2}{3}}}dx \qquad\qquad u = x - 1 \to du = dx$$

$$\int \frac{1}{(x-1)^{\frac{2}{3}}}dx = \int \frac{1}{(u)^{\frac{2}{3}}}dx = \int u^{-\frac{2}{3}}\,dx = \frac{u^{-\frac{2}{3}+1}}{-\frac{2}{3}+1} = \frac{u^{\frac{1}{3}}}{\frac{1}{3}} = u^{\frac{1}{3}} \cdot \frac{3}{1} = 3u^{\frac{1}{3}} = 3(x-1)^{\frac{1}{3}} + C$$

$$I_1 = lim_{t\to 1}\int_0^t \frac{1}{(x-1)^{\frac{2}{3}}}\,dx = lim_{t\to 1}\left[3(x-1)^{\frac{1}{3}}\right]_0^t = lim_{t\to 1}3(t-1)^{\frac{1}{3}} - 3(0-1)^{\frac{1}{3}}$$

$$= 0 - 3(-1) = 0 + 3 = 3$$

$$I_2 = lim_{t\to 1}\int_t^2 \frac{1}{(x-1)^{\frac{2}{3}}}dx = lim_{t\to 1}\left[3(x-1)^{\frac{1}{3}}\right]_t^2 = lim_{t\to 1}3(2-1)^{\frac{1}{3}} - 3(t-1)^{\frac{1}{3}}$$

$$= 3(1) - 0 = 3 - 0 = 3$$

$$I = I_1 + I_2 = 3 + 3 = 6$$

Therefore, the integral is convergent and converges to 6.

6 Marks

7. Solve the following differential equation. Solve explicitly for *y(x)*.

$$\left(x^2 + 3x + 2\right)y' = y$$

$$(x^2 + 3x + 2)\frac{dy}{dx} = y \Rightarrow \frac{dy}{dx} = \frac{y}{(x^2+3x+2)} \Rightarrow \frac{1}{y}\,dy = \frac{1}{(x^2+3x+2)}dx$$

$$\int \frac{1}{y}\,dy = \int \frac{1}{(x^2+3x+2)}dx\,, \qquad \int \frac{1}{(x^2+3x+2)}dx = \ ?$$

$$\int \frac{1}{(x^2+3x+2)}dx$$

$$x^2 + 3x + 2 = 0 \qquad p = 2 \qquad s = 3: \qquad 1\ \&\ 2$$

$$x^2 + x + 2x + 2 = 0 \Rightarrow x(x + 1) + 2(x + 1) = 0 \Rightarrow (x + 1)(x + 2) = 0$$

$$x =- 1,\ x =- 2$$

$$\frac{1}{(x^2+3x+2)} = \frac{1}{(x+1)(x+2)} = \frac{A}{(x+1)} + \frac{B}{(x+2)} \Rightarrow 1 = A(x + 2) + B(x + 1)$$

$x =- 1$: $\quad 1 = A(- 1 + 2) + B(- 1 + 1)$

$1 = A(1) + 0 \Rightarrow A = 1$

$x =- 2$: $\quad 1 = A(- 2 + 2) + B(- 2 + 1)$

$1 = 0 + B(- 1) \Rightarrow B =- 1$

$$\frac{1}{(x+1)(x+2)} = \frac{A}{(x+1)} + \frac{B}{(x+2)} = \frac{1}{(x+1)} - \frac{1}{(x+2)}$$

$$\int \frac{1}{(x+1)} - \frac{1}{(x+2)}dx = \int \frac{1}{(x+1)}dx - \int \frac{1}{(x+2)}dx = ln|x + 1| - ln|x + 2| + C$$

$$\int \frac{1}{y}\,dy = \int \frac{1}{(x^2+3x+2)}dx \Rightarrow ln|y| = ln|x + 1| - ln|x + 2| + C$$

$$e^{[ln|y|=ln|x+1|-ln|x+2|+C]} \Rightarrow e^{ln|y|} = e^{ln|x+1|} \cdot e^{-ln|(x+2)|} \cdot e^{C} \Rightarrow e^{ln|y|} = e^{ln|x+1|} \cdot e^{ln|(x+2)^{-1}|} \cdot e^{C}$$

$$|y| = |x + 1| \cdot |(x + 2)^{-1}| \cdot C = \frac{x+1}{x+2} \cdot C$$

Therefore, $y(x) = \frac{x+1}{x+2} \cdot C$.

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